Arduino UNO R4 WiFi CAN Bus

Learn how to send messages using the CAN bus on the UNO R4 WiFi.

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In this tutorial you will learn how to use the CAN controller on the **Arduino UNO R4 WiFi** board. The CAN controller is embedded in the UNO R4 WiFi's microcontroller (RA4M1). CAN is a serial protocol that is mainly used in the automotive industry.

Please note that CAN controller requires an external transceiver to function. Instructions and hardware examples are provided in this tutorial.

Goals

The goals of this tutorial are:

- Get an overview of the built-in CAN library
- Learn how to connect a board to a CAN transceiver module
- Send a CAN message between two Arduinos

Help

Hardware & Software Needed

- Arduino IDE (online or offline)
- Arduino R4 WiFi
- UNO R4 Board Package
- CAN transceiver module *
- Jumper wires

* In this tutorial, we are using a SN65HVD230 breakout module.

Controller Area Network (CAN)

The CAN bus uses two wires: **CAN high** and **CAN low**. On the UNO R4 WiFi, these pins are:

- D13/CANRX0 (receive)
- D10/CANTX0 (transmit)

To communicate with other CAN devices however, you need a transceiver module. In this tutorial, we will be using a SN65HVD230 breakout. To connect this, you can follow the circuit diagram available in the section below.

For this tutorial, we will use a simple example that sends a CAN message between two UNO R4 WiFi devices. If you wish, you can also connect an existing CAN device to the UNO R4 WiFi.

Circuit

To connect the CAN transceiver, follow the table and circuit diagram below:

UNO R4 WiFi	CAN Transceiver
D13 (CANRX0)	CANRX
D10 (CANTX0)	CANTX
3.3V	VIN
GND	GND

Then, between the CAN transceivers, connect the following:

CAN Transceiver 1	CAN Transceiver 2
CANH (HIGH)	CANH (HIGH)
CANL (LOW)	CANL (LOW)

Code Examples

The following code examples need to be uploaded to each of the UNO R4 WiFi boards, one will send a message, one will receive it. These examples are available in the UNO R4 Board Package, and using the Arduino IDE, you can access them by navigating to **File > Examples > Arduino_CAN > CANWrite/CANRead**

The library used is built into the Board Package, so no need to install the library if you have the Board Package installed.

To initialize the library, use `CAN.begin(CanBitRate::BR_250k)`, where a CAN bit rate is specified. Choose between:

- BR_125k (125000)
- BR_250k (250000)
- BR_500k (500000)

CAN Write

To send a CAN message, you can create a `CanMsg` object, which should contain the `CAN_ID`, size and message data. Below is an example on how to create such object.

```
uint8_t const msg_data[
memcpy((void *)(msg_dat
CanMsg msg(CAN_ID, size
```

After you have crafted a CAN message, we can send it off, by using the `CAN.write()` method. The following example creates a CAN message that increases each time `void loop()` is executed.

Code Source on Github

```
/*
  CANWrite

  Write and send CAN

  See the full docume
  https://docs.arduin
*/

/*******************
 * INCLUDE
 *******************

#include <Arduino_CAN

/*******************
 * CONSTANTS
 *******************

static uint32_t const

/*******************
 * SETUP/LOOP
 *******************

void setup()
{
  Serial.begin(115200
```

CAN Read

To read an incoming CAN message, first use `CAN.available()` to check if data is available, before using `CAN.read()` to read the message.

Code Source on Github

```
/*
  CANRead

  Receive and read CA

  See the full docume
  https://docs.arduin
*/

/*******************
 * INCLUDE
 *******************

#include <Arduino_CAN

/*******************
 * SETUP/LOOP
 *******************

void setup()
{
  Serial.begin(115200
  while (!Serial) { }

  if (!CAN.begin(CanB
  {
    Serial.println("C
    for (;;) {}
```

Summary

This tutorial shows how to use the CAN bus available on the UNO R4 WiFi, and how to send and receive data using the Arduino_CAN library.

Read more about this board in the [Arduino UNO R4 WiFi documentation](#).

through a public GitHub repository. If you see anything wrong, you can edit this page here.

Discord

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